RESEARCH POSTER ABSTRACT

Comparing Software Process Models in Cloud-Native and Live-Service Architectures: An Empirical Analysis of Netflix and Riot Games

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ABSTRACT

Software process models structure how systems are specified, designed, validated, and evolved. This study presents a comparative software engineering analysis of two distinct large-scale industry systems: Netflix's cloud-native video streaming platform and Riot Games' live-service multiplayer game engine (League of Legends). Software development is analyzed across four core process activities—requirements specification, design and implementation, verification and validation (V&V), and continuous evolution—contrasting plan-driven Waterfall, incremental development, integration and configuration, and Agile methodologies. Netflix utilizes a fully continuous, microservices-driven DevOps model that executes thousands of production deployments daily, managed through feature flags, dynamic A/B telemetry, and Chaos Engineering fault injection. In contrast, Riot Games employs a hybrid Agile-incremental framework governed by tightly synchronized, bi-weekly global patch cadences, multi-tier public beta playtesting (PBE), and deterministic client-server validation to protect competitive integrity. The comparison demonstrates that software process selection is fundamentally constrained by system architecture, team topology, safety/security risk, and required release frequency rather than adherence to a universal "best" methodology. Large-scale platforms naturally converge toward hybrid lifecycles that combine plan-driven architectural baselines with iterative sprint execution. Emerging considerations, including AI-assisted static analysis and automated canary verification, further reinforce the principle that process models must dynamically fit operational risk and domain constraints.

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KEYWORDS

Software Process Models; Agile Development; Incremental Delivery; Waterfall; Chaos Engineering; Microservices; Live-Service Games; Netflix; Riot Games.